KITABU QUEST

Grade 11

AGRICULTURE

Honeybee

Cover scene

learners working in a school garden with seedlings, compost, watering cans, and a friendly honeybee mascot

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE
11

KEN CBC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST Grade 11 Agriculture

Kenya CBC content draft for review

Mascot: honeybee

This content-first draft was regenerated from Kitabu curriculum data using a topic-by-topic book plan. It is designed to help learners read, practise, discuss, create, and revise with confidence.

Cover artwork is intentionally deferred until all content has passed review and is marked ready-for-cover.

How To Use This Book

This book helps Grade 11 learners study Agriculture through clear lessons, worked examples, activities, practice, and revision.

Use the inquiry questions to think first. Read the explanation. Try the guided example. Complete the activity. Answer the practice questions. Correct your work and ask for support where needed.

Learn through school-garden practice, observation logs, tool safety, and home projects.

Every unit has a local example, a misconception to avoid, success criteria, and a check for understanding.

Learning Skills In This Book

Each chapter builds knowledge, skills, values, creativity, communication, collaboration, critical thinking, digital literacy, and self-confidence.

Look for these page types: Learn, Example, Activity, Practice, Reflection, Home Link, and Review.

How to study well: read the goal first, underline new words, try the example before reading the answer, discuss with a partner, and correct your work after feedback. Do not skip the reflection questions. They help you notice what you understand and what needs more practice.

Table Of Contents

Use this table of contents to plan your study. Start with the first chapter, but return to earlier pages whenever you need revision.

After each chapter, complete the chapter review before moving forward. At the end of the book, use the glossary, answer notes, and final project to revise the whole subject.

Chapter: Crop Production

In this chapter you will study Crop Production.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:
- 1.1 Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment
- 1.2 Water conservation Methods of irrigation Importance of irrigation
- 1.3 Vegetative propagation Methods of vegetative propagation Use of vegetative propagation
- 1.4 Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry
- 1.5 Crop Protection: Pest control Categories of pests Control of pests Economic importance of pests
- 1.6 Crop Production Structures in crop production Construction materials Construction of structures

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Lesson Opener

Learning area: Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients.

By the end of this lesson sequence, you should be able to:
- describe the role of nutrients in healthy crop growth
- categorize inorganic fertilisers used in crop production
- carry out soil sampling for analysis
- apply inorganic fertiliser for crop production

Inquiry questions:
- How do inorganic fertilisers influence crop production?

Before reading, write one thing you already know and one question you want this lesson to answer.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients.

The important idea is Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients
- Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment
- Inorganic
- fertilisers
- nutrients
- Categories
- sampling
- amendment

Explain the idea in your own words before attempting the activities.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Worked Example

Practical model for Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Soil observation and improvement.

Procedure:
1. Collect a small soil sample from an approved school-garden spot.
2. Observe colour, texture, moisture, stones, plant remains, and signs of erosion.
3. Record the evidence in a table before suggesting an improvement.
4. Choose one improvement such as mulching, compost, contour planting, cover crops, or safe drainage.
5. Explain how the improvement protects soil fertility, water, or plant growth.

Hazards: sharp objects, dirty hands, unsafe slopes, and contaminated soil.

Rubric: excellent work names the soil evidence, links it to a practical improvement, includes safety, and records observations clearly.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Activity

Activity for Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Complete a supervised agriculture procedure card.

Inquiry question: How do inorganic fertilisers influence crop production?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Activity And Practice

Practice for Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients:

1. Carry out or describe a practical task for this outcome: describe the role of nutrients in healthy crop growth. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: categorize inorganic fertilisers used in crop production. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: carry out soil sampling for analysis. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: apply inorganic fertiliser for crop production. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to inorganic fertilisers soil nutrients categories of fertilisers soil sampling soil ph amendment: role of nutrients and write two useful sentences about it.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Outcome Check

Outcome: describe the role of nutrients in healthy crop growth

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Outcome Check

Outcome: categorize inorganic fertilisers used in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients: Outcome Check

Outcome: carry out soil sampling for analysis

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: role of nutrients.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices: Lesson Opener

Learning area: Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices.

By the end of this lesson sequence, you should be able to:
- propose soil amendment practices for crop production
- acknowledge the importance of inorganic fertilisers in crop production.

Inquiry questions:
- How do inorganic fertilisers influence crop production?

Before reading, write one thing you already know and one question you want this lesson to answer.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices.

The important idea is Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices
- Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment
- Inorganic
- fertilisers
- nutrients
- Categories
- sampling
- amendment

Explain the idea in your own words before attempting the activities.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices: Worked Example

Practical model for Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices: Soil observation and improvement.

Procedure:
1. Collect a small soil sample from an approved school-garden spot.
2. Observe colour, texture, moisture, stones, plant remains, and signs of erosion.
3. Record the evidence in a table before suggesting an improvement.
4. Choose one improvement such as mulching, compost, contour planting, cover crops, or safe drainage.
5. Explain how the improvement protects soil fertility, water, or plant growth.

Hazards: sharp objects, dirty hands, unsafe slopes, and contaminated soil.

Rubric: excellent work names the soil evidence, links it to a practical improvement, includes safety, and records observations clearly.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices: Activity

Activity for Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices: Complete a supervised agriculture procedure card.

Inquiry question: How do inorganic fertilisers influence crop production?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices: Activity And Practice

Practice for Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices:

1. Carry out or describe a practical task for this outcome: propose soil amendment practices for crop production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: acknowledge the importance of inorganic fertilisers in crop production.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to inorganic fertilisers soil nutrients categories of fertilisers soil sampling soil ph amendment: propose soil amendment practices and write two useful sentences about it.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices: Outcome Check

Outcome: propose soil amendment practices for crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices: Outcome Check

Outcome: acknowledge the importance of inorganic fertilisers in crop production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: propose soil amendment practices.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Water conservation Methods of irrigation Importance of irrigation: Lesson Opener

Learning area: Water conservation Methods of irrigation Importance of irrigation

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Water conservation Methods of irrigation Importance of irrigation.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Water conservation Methods of irrigation Importance of irrigation.

By the end of this lesson sequence, you should be able to:
- describe methods of irrigation in crop production
- apply efficient approaches to irrigation in crop production
- appreciate the importance of irrigation in crop production.

Inquiry questions:
- How does efficient irrigation of crops conserve water?

Before reading, write one thing you already know and one question you want this lesson to answer.

Water conservation Methods of irrigation Importance of irrigation: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Water conservation Methods of irrigation Importance of irrigation.

The important idea is Water conservation Methods of irrigation Importance of irrigation. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Water conservation Methods of irrigation Importance of irrigation.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Water conservation Methods of irrigation Importance of irrigation
- Water
- conservation
- Methods
- irrigation

Explain the idea in your own words before attempting the activities.

Water conservation Methods of irrigation Importance of irrigation: Worked Example

Practical model for Water conservation Methods of irrigation Importance of irrigation: Crop-care decision card.

Procedure:
1. Identify the crop stage: seed, nursery, transplanted crop, flowering, fruiting, or harvest.
2. Observe spacing, water, weeds, pests, disease signs, soil cover, and plant colour.
3. Choose one safe action: watering, thinning, weeding, mulching, staking, pest scouting, or harvesting.
4. Record why that action fits the evidence seen.
5. Review the result after a set time and note whether plant health improved.

Hazards: sharp tools, unsafe chemicals, dirty water, plant irritation, and overwatering.

Rubric: excellent work uses real observations, names the crop-care action, protects safety, and explains the expected result.

Water conservation Methods of irrigation Importance of irrigation: Activity

Activity for Water conservation Methods of irrigation Importance of irrigation: Complete a supervised agriculture procedure card.

Inquiry question: How does efficient irrigation of crops conserve water?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Water conservation Methods of irrigation Importance of irrigation: Activity And Practice

Practice for Water conservation Methods of irrigation Importance of irrigation:

1. Carry out or describe a practical task for this outcome: describe methods of irrigation in crop production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: apply efficient approaches to irrigation in crop production. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: appreciate the importance of irrigation in crop production.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to water conservation methods of irrigation importance of irrigation and write two useful sentences about it.

Water conservation Methods of irrigation Importance of irrigation: Outcome Check

Outcome: describe methods of irrigation in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Water conservation Methods of irrigation Importance of irrigation.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Water conservation Methods of irrigation Importance of irrigation: Outcome Check

Outcome: apply efficient approaches to irrigation in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Water conservation Methods of irrigation Importance of irrigation.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Water conservation Methods of irrigation Importance of irrigation: Outcome Check

Outcome: appreciate the importance of irrigation in crop production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Water conservation Methods of irrigation Importance of irrigation.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Lesson Opener

Learning area: Vegetative propagation Methods of vegetative propagation Use of vegetative propagation

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Vegetative propagation Methods of vegetative propagation Use of vegetative propagation.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Vegetative propagation Methods of vegetative propagation Use of vegetative propagation.

By the end of this lesson sequence, you should be able to:
- describe the methods of vegetative propagation in crop production
- carry out vegetative propagation in crop production
- appraise the use of vegetative propagation in crop production.

Inquiry questions:
- How is vegetative propagation carried out?

Before reading, write one thing you already know and one question you want this lesson to answer.

Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Vegetative propagation Methods of vegetative propagation Use of vegetative propagation.

The important idea is Vegetative propagation Methods of vegetative propagation Use of vegetative propagation. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Vegetative propagation Methods of vegetative propagation Use of vegetative propagation.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Vegetative propagation Methods of vegetative propagation Use of vegetative propagation
- Vegetative
- propagation
- Methods

Explain the idea in your own words before attempting the activities.

Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Worked Example

Practical model for Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Crop-care decision card.

Procedure:
1. Identify the crop stage: seed, nursery, transplanted crop, flowering, fruiting, or harvest.
2. Observe spacing, water, weeds, pests, disease signs, soil cover, and plant colour.
3. Choose one safe action: watering, thinning, weeding, mulching, staking, pest scouting, or harvesting.
4. Record why that action fits the evidence seen.
5. Review the result after a set time and note whether plant health improved.

Hazards: sharp tools, unsafe chemicals, dirty water, plant irritation, and overwatering.

Rubric: excellent work uses real observations, names the crop-care action, protects safety, and explains the expected result.

Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Activity

Activity for Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Complete a supervised agriculture procedure card.

Inquiry question: How is vegetative propagation carried out?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Activity And Practice

Practice for Vegetative propagation Methods of vegetative propagation Use of vegetative propagation:

1. Carry out or describe a practical task for this outcome: describe the methods of vegetative propagation in crop production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: carry out vegetative propagation in crop production. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: appraise the use of vegetative propagation in crop production.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to vegetative propagation methods of vegetative propagation use of vegetative propagation and write two useful sentences about it.

Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Outcome Check

Outcome: describe the methods of vegetative propagation in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Vegetative propagation Methods of vegetative propagation Use of vegetative propagation.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Outcome Check

Outcome: carry out vegetative propagation in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Vegetative propagation Methods of vegetative propagation Use of vegetative propagation.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: Outcome Check

Outcome: appraise the use of vegetative propagation in crop production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Vegetative propagation Methods of vegetative propagation Use of vegetative propagation.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Lesson Opener

Learning area: Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry.

By the end of this lesson sequence, you should be able to:
- examine the characteristics of agro- forestry trees for conservation of the environment
- describe natural regeneration approach in environmental restoration
- grow agroforestry trees for conservation of the environment
- appreciate the importance of agro-forestry in conserving the environment.

Inquiry questions:
- Why should we practice agroforestry?
- How do agroforestry trees conserve environment?

Before reading, write one thing you already know and one question you want this lesson to answer.

Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry.

The important idea is Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry
- Agroforestry
- Characteristics
- Natural
- regeneration
- trees

Explain the idea in your own words before attempting the activities.

Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Worked Example

Practical model for Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Water-use plan for a garden bed.

Procedure:
1. Check soil moisture before watering by observing colour and feel.
2. Decide whether the bed needs watering, mulching, drainage, or shade.
3. Water near the root zone and avoid wasting water on paths.
4. Record amount, time, weather, and plant response.
5. Suggest one conservation action such as mulching, watering early, fixing leaks, or using a watering can with a rose.

Hazards: slippery paths, contaminated water, heavy containers, and flooded roots.

Rubric: excellent work links water action to evidence, avoids waste, protects plants, and keeps a useful record.

Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Activity

Activity for Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Complete a supervised agriculture procedure card.

Inquiry question: Why should we practice agroforestry?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Activity And Practice

Practice for Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry:

1. Carry out or describe a practical task for this outcome: examine the characteristics of agro- forestry trees for conservation of the environment. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: describe natural regeneration approach in environmental restoration. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: grow agroforestry trees for conservation of the environment. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: appreciate the importance of agro-forestry in conserving the environment.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to agroforestry characteristics of agroforestry natural regeneration agroforestry trees importance of agroforestry and write two useful sentences about it.

Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Outcome Check

Outcome: examine the characteristics of agro- forestry trees for conservation of the environment

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Outcome Check

Outcome: describe natural regeneration approach in environmental restoration

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry: Outcome Check

Outcome: grow agroforestry trees for conservation of the environment

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agroforestry Characteristics of Agroforestry Natural regeneration Agroforestry trees Importance of agroforestry.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Protection: Pest control Categories of pests Control of pests Economic importance: Lesson Opener

Learning area: Crop Protection: Pest control Categories of pests Control of pests Economic importance

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Crop Protection: Pest control Categories of pests Control of pests Economic importance.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Crop Protection: Pest control Categories of pests Control of pests Economic importance.

By the end of this lesson sequence, you should be able to:
- examine categories of pests in crop production
- control pests in crop production
- appreciate the economic importance of pests in crop production.

Inquiry questions:
- How are pests important in crop production?
- How can pests be controlled in crop production?

Before reading, write one thing you already know and one question you want this lesson to answer.

Crop Protection: Pest control Categories of pests Control of pests Economic importance: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Crop Protection: Pest control Categories of pests Control of pests Economic importance.

The important idea is Crop Protection: Pest control Categories of pests Control of pests Economic importance. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Crop Protection: Pest control Categories of pests Control of pests Economic importance.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Crop Protection: Pest control Categories of pests Control of pests Economic importance
- Crop Protection: Pest control Categories of pests Control of pests Economic importance of pests
- Protection
- control
- Categories
- pests
- Economic

Explain the idea in your own words before attempting the activities.

Crop Protection: Pest control Categories of pests Control of pests Economic importance: Worked Example

Practical model for Crop Protection: Pest control Categories of pests Control of pests Economic importance: Crop-care decision card.

Procedure:
1. Identify the crop stage: seed, nursery, transplanted crop, flowering, fruiting, or harvest.
2. Observe spacing, water, weeds, pests, disease signs, soil cover, and plant colour.
3. Choose one safe action: watering, thinning, weeding, mulching, staking, pest scouting, or harvesting.
4. Record why that action fits the evidence seen.
5. Review the result after a set time and note whether plant health improved.

Hazards: sharp tools, unsafe chemicals, dirty water, plant irritation, and overwatering.

Rubric: excellent work uses real observations, names the crop-care action, protects safety, and explains the expected result.

Crop Protection: Pest control Categories of pests Control of pests Economic importance: Activity

Activity for Crop Protection: Pest control Categories of pests Control of pests Economic importance: Complete a supervised agriculture procedure card.

Inquiry question: How are pests important in crop production?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Crop Protection: Pest control Categories of pests Control of pests Economic importance: Activity And Practice

Practice for Crop Protection: Pest control Categories of pests Control of pests Economic importance:

1. Carry out or describe a practical task for this outcome: examine categories of pests in crop production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: control pests in crop production. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: appreciate the economic importance of pests in crop production.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to crop protection: pest control categories of pests control of pests economic importance and write two useful sentences about it.

Crop Protection: Pest control Categories of pests Control of pests Economic importance: Outcome Check

Outcome: examine categories of pests in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Crop Protection: Pest control Categories of pests Control of pests Economic importance.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Protection: Pest control Categories of pests Control of pests Economic importance: Outcome Check

Outcome: control pests in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Crop Protection: Pest control Categories of pests Control of pests Economic importance.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Protection: Pest control Categories of pests Control of pests Economic importance: Outcome Check

Outcome: appreciate the economic importance of pests in crop production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Crop Protection: Pest control Categories of pests Control of pests Economic importance.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Production Structures in crop production Construction materials Construction of structures: Lesson Opener

Learning area: Crop Production Structures in crop production Construction materials Construction of structures

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Crop Production Structures in crop production Construction materials Construction of structures.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Crop Production Structures in crop production Construction materials Construction of structures.

By the end of this lesson sequence, you should be able to:
- describe various structures used in crop production
- propose appropriate materials for construction of a crop production structure
- construct a structure for a specified crop production purpose
- adopt the use of crop production structures in various production practices.

Inquiry questions:
- How can structural facilities enhance crop production processes?

Before reading, write one thing you already know and one question you want this lesson to answer.

Crop Production Structures in crop production Construction materials Construction of structures: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Crop Production Structures in crop production Construction materials Construction of structures.

The important idea is Crop Production Structures in crop production Construction materials Construction of structures. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Crop Production Structures in crop production Construction materials Construction of structures.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Crop Production Structures in crop production Construction materials Construction of structures
- Production
- Structures
- Construction
- materials

Explain the idea in your own words before attempting the activities.

Crop Production Structures in crop production Construction materials Construction of structures: Worked Example

Practical model for Crop Production Structures in crop production Construction materials Construction of structures: Crop-care decision card.

Procedure:
1. Identify the crop stage: seed, nursery, transplanted crop, flowering, fruiting, or harvest.
2. Observe spacing, water, weeds, pests, disease signs, soil cover, and plant colour.
3. Choose one safe action: watering, thinning, weeding, mulching, staking, pest scouting, or harvesting.
4. Record why that action fits the evidence seen.
5. Review the result after a set time and note whether plant health improved.

Hazards: sharp tools, unsafe chemicals, dirty water, plant irritation, and overwatering.

Rubric: excellent work uses real observations, names the crop-care action, protects safety, and explains the expected result.

Crop Production Structures in crop production Construction materials Construction of structures: Activity

Activity for Crop Production Structures in crop production Construction materials Construction of structures: Complete a supervised agriculture procedure card.

Inquiry question: How can structural facilities enhance crop production processes?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Crop Production Structures in crop production Construction materials Construction of structures: Activity And Practice

Practice for Crop Production Structures in crop production Construction materials Construction of structures:

1. Carry out or describe a practical task for this outcome: describe various structures used in crop production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: propose appropriate materials for construction of a crop production structure. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: construct a structure for a specified crop production purpose. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: adopt the use of crop production structures in various production practices.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to crop production structures in crop production construction materials construction of structures and write two useful sentences about it.

Crop Production Structures in crop production Construction materials Construction of structures: Outcome Check

Outcome: describe various structures used in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Crop Production Structures in crop production Construction materials Construction of structures.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Production Structures in crop production Construction materials Construction of structures: Outcome Check

Outcome: propose appropriate materials for construction of a crop production structure

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Crop Production Structures in crop production Construction materials Construction of structures.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Production Structures in crop production Construction materials Construction of structures: Outcome Check

Outcome: construct a structure for a specified crop production purpose

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Crop Production Structures in crop production Construction materials Construction of structures.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Production: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Animal Production

In this chapter you will study Animal Production.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:
- 2.1 Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry digestive system
- 2.2 Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of parasites Harmful effects
- 2.3 Structures in Animal production Structures in livestock production Siting of structures Construction of structures Importance
- 2.4 Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish farming.
- 2.5 Rearing Selected Animals Management practices
- 2.6 Livestock Products Obtaining products: wool shearing Obtaining products: Milking

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry: Lesson Opener

Learning area: Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry.

By the end of this lesson sequence, you should be able to:
- describe digestive systems of animals
- relate the digestive compare how effective ruminant animals systems of animals to appropriate type of animal
- appreciate diversity of digestive systems of animals to their productivity.

Inquiry questions:
- How do farm animals extract nutrients from feeds?
- Why are various farm animals fed on different types of feed?

Before reading, write one thing you already know and one question you want this lesson to answer.

Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry.

The important idea is Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry
- Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry digestive system
- Digestive
- systems
- animals
- Ruminants
- system
- Poultry

Explain the idea in your own words before attempting the activities.

Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry: Worked Example

Practical model for Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry: Safe animal handling.

Teacher-supervised procedure:
1. Observe the animal's behaviour before moving close.
2. Approach from the side where the animal can see you.
3. Keep hands away from the mouth, horns, hooves, and back legs.
4. Use clean water, feed, bedding, and equipment as instructed.
5. Wash hands and disinfect equipment after the task.

Hazards: kicks, bites, scratches, disease transmission, and stress to the animal.

Quality criteria: calm handling, clean tools, correct restraint if needed, animal welfare, and accurate records of feed, health, or behaviour.

Digestive systems of animals Ruminants digestive system Non-ruminants digestive system Poultry: Activity

Activity for Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry: Complete a supervised agriculture procedure card.

Inquiry question: How do farm animals extract nutrients from feeds?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Digestive systems of animals Ruminants digestive system Non-ruminants digestive system Poultry: Activity And Practice

Practice for Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry:

1. Carry out or describe a practical task for this outcome: describe digestive systems of animals. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: relate the digestive compare how effective ruminant animals systems of animals to appropriate type of animal. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: appreciate diversity of digestive systems of animals to their productivity.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to digestive systems of animals ruminants digestive system non- ruminants digestive system poultry and write two useful sentences about it.

Digestive systems of animals Ruminants digestive system Non-ruminants digestive system Poultry: Outcome Check

Outcome: describe digestive systems of animals

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Digestive systems of animals Ruminants digestive system Non-ruminants digestive system Poultry: Outcome Check

Outcome: relate the digestive compare how effective ruminant animals systems of animals to appropriate type of animal

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Digestive systems of animals Ruminants digestive system Non-ruminants digestive system Poultry: Outcome Check

Outcome: appreciate diversity of digestive systems of animals to their productivity.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Digestive systems of animals Ruminants digestive system Non- ruminants digestive system Poultry.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control: Lesson Opener

Learning area: Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control.

By the end of this lesson sequence, you should be able to:
- describe parasites affecting animal production
- illustrate the lifecycle of parasites in animal production
- control parasites in animal production
- acknowledge the harmful effects of parasites in animal production.

Inquiry questions:
- How do parasites affect animal production?
- How are parasites controlled in animal production?

Before reading, write one thing you already know and one question you want this lesson to answer.

Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of.

The important idea is Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control
- Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of parasites Harmful effects
- Animal
- Health
- Parasite
- control
- Categories
- parasites

Explain the idea in your own words before attempting the activities.

Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control: Worked Example

Practical model for Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of: Choose the exact agriculture action, then prove it with evidence.

Procedure:
1. Choose one concrete task from this topic: observe soil, care for a crop, handle a tool, feed or protect an animal, store produce, manage water, or keep a record.
2. State the purpose of that task in one sentence.
3. List exact tools, materials, living things, and safety items before starting.
4. Perform one step at a time under teacher or adult guidance where needed.
5. Record date, condition observed, action taken, result, and one improvement.

Rubric: excellent work names the real task, uses safe handling, gives a clear sequence, cares for living things, and records useful evidence.

Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control: Activity

Activity for Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of: Complete a supervised agriculture procedure card.

Inquiry question: How do parasites affect animal production?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control: Activity And Practice

Practice for Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control:

1. Carry out or describe a practical task for this outcome: describe parasites affecting animal production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: illustrate the lifecycle of parasites in animal production. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: control parasites in animal production. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: acknowledge the harmful effects of parasites in animal production.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to animal health: parasite control categories of parasites lifecycle of parasites control and write two useful sentences about it.

Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control: Outcome Check

Outcome: describe parasites affecting animal production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control: Outcome Check

Outcome: illustrate the lifecycle of parasites in animal production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control: Outcome Check

Outcome: control parasites in animal production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Health: Parasite control Categories of parasites Lifecycle of parasites Control of.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Structures in Animal production Structures in livestock production Siting of structures Construction: Lesson Opener

Learning area: Structures in Animal production Structures in livestock production Siting of structures Construction

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Structures in Animal production Structures in livestock production Siting of structures Construction.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Structures in Animal production Structures in livestock production Siting of structures Construction.

By the end of this lesson sequence, you should be able to:
- describe structures used in livestock production
- analyse factors to consider in siting livestock production structures
- construct a structure for specified animal production purpose
- appreciate the importance of various structures in animal production.

Inquiry questions:
- How are farm structures used in livestock production processes?

Before reading, write one thing you already know and one question you want this lesson to answer.

Structures in Animal production Structures in livestock production Siting of structures Construction: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Structures in Animal production Structures in livestock production Siting of structures Construction.

The important idea is Structures in Animal production Structures in livestock production Siting of structures Construction. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Structures in Animal production Structures in livestock production Siting of structures Construction.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Structures in Animal production Structures in livestock production Siting of structures Construction
- Structures in Animal production Structures in livestock production Siting of structures Construction of structures Importance
- Structures
- Animal
- production
- livestock
- Siting
- Construction

Explain the idea in your own words before attempting the activities.

Structures in Animal production Structures in livestock production Siting of structures Construction: Worked Example

Practical model for Structures in Animal production Structures in livestock production Siting of structures Construction: Safe animal handling.

Teacher-supervised procedure:
1. Observe the animal's behaviour before moving close.
2. Approach from the side where the animal can see you.
3. Keep hands away from the mouth, horns, hooves, and back legs.
4. Use clean water, feed, bedding, and equipment as instructed.
5. Wash hands and disinfect equipment after the task.

Hazards: kicks, bites, scratches, disease transmission, and stress to the animal.

Quality criteria: calm handling, clean tools, correct restraint if needed, animal welfare, and accurate records of feed, health, or behaviour.

Structures in Animal production Structures in livestock production Siting of structures Construction: Activity

Activity for Structures in Animal production Structures in livestock production Siting of structures Construction: Complete a supervised agriculture procedure card.

Inquiry question: How are farm structures used in livestock production processes?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Structures in Animal production Structures in livestock production Siting of structures Construction: Activity And Practice

Practice for Structures in Animal production Structures in livestock production Siting of structures Construction:

1. Carry out or describe a practical task for this outcome: describe structures used in livestock production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: analyse factors to consider in siting livestock production structures. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: construct a structure for specified animal production purpose. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: appreciate the importance of various structures in animal production.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to structures in animal production structures in livestock production siting of structures construction and write two useful sentences about it.

Structures in Animal production Structures in livestock production Siting of structures Construction: Outcome Check

Outcome: describe structures used in livestock production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Structures in Animal production Structures in livestock production Siting of structures Construction.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Structures in Animal production Structures in livestock production Siting of structures Construction: Outcome Check

Outcome: analyse factors to consider in siting livestock production structures

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Structures in Animal production Structures in livestock production Siting of structures Construction.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Structures in Animal production Structures in livestock production Siting of structures Construction: Outcome Check

Outcome: construct a structure for specified animal production purpose

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Structures in Animal production Structures in livestock production Siting of structures Construction.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Lesson Opener

Learning area: Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish.

By the end of this lesson sequence, you should be able to:
- describe fish farming in animal production
- carry out management practices in fish rearing
- appreciate innovative approaches to fish farming.

Inquiry questions:
- How are fish reared?

Before reading, write one thing you already know and one question you want this lesson to answer.

Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish.

The important idea is Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish
- Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish farming.
- Aquaculture
- farming
- Management
- practices
- Innovative
- approaches

Explain the idea in your own words before attempting the activities.

Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Worked Example

Practical model for Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Choose the exact agriculture action, then prove it with evidence.

Procedure:
1. Choose one concrete task from this topic: observe soil, care for a crop, handle a tool, feed or protect an animal, store produce, manage water, or keep a record.
2. State the purpose of that task in one sentence.
3. List exact tools, materials, living things, and safety items before starting.
4. Perform one step at a time under teacher or adult guidance where needed.
5. Record date, condition observed, action taken, result, and one improvement.

Rubric: excellent work names the real task, uses safe handling, gives a clear sequence, cares for living things, and records useful evidence.

Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Activity

Activity for Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Complete a supervised agriculture procedure card.

Inquiry question: How are fish reared?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Activity And Practice

Practice for Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish:

1. Carry out or describe a practical task for this outcome: describe fish farming in animal production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: carry out management practices in fish rearing. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: appreciate innovative approaches to fish farming.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to aquaculture fish farming management practices in fish farming innovative approaches to fish and write two useful sentences about it.

Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Outcome Check

Outcome: describe fish farming in animal production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Outcome Check

Outcome: carry out management practices in fish rearing

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish: Outcome Check

Outcome: appreciate innovative approaches to fish farming.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Aquaculture Fish farming Management practices in fish farming Innovative approaches to fish.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Rearing Selected Animals Management practices: Lesson Opener

Learning area: Rearing Selected Animals Management practices

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Rearing Selected Animals Management practices.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Rearing Selected Animals Management practices.

By the end of this lesson sequence, you should be able to:
- describe the management practices in animal rearing
- carry out routine management practices in animal rearing
- develop responsibility in the rearing of selected animals.

Inquiry questions:
- How do we rear domestic animals?

Before reading, write one thing you already know and one question you want this lesson to answer.

Rearing Selected Animals Management practices: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Rearing Selected Animals Management practices.

The important idea is Rearing Selected Animals Management practices. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Rearing Selected Animals Management practices.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Rearing Selected Animals Management practices
- Rearing
- Animals
- Management
- practices

Explain the idea in your own words before attempting the activities.

Rearing Selected Animals Management practices: Worked Example

Practical model for Rearing Selected Animals Management practices: Safe animal handling.

Teacher-supervised procedure:
1. Observe the animal's behaviour before moving close.
2. Approach from the side where the animal can see you.
3. Keep hands away from the mouth, horns, hooves, and back legs.
4. Use clean water, feed, bedding, and equipment as instructed.
5. Wash hands and disinfect equipment after the task.

Hazards: kicks, bites, scratches, disease transmission, and stress to the animal.

Quality criteria: calm handling, clean tools, correct restraint if needed, animal welfare, and accurate records of feed, health, or behaviour.

Rearing Selected Animals Management practices: Activity

Activity for Rearing Selected Animals Management practices: Complete a supervised agriculture procedure card.

Inquiry question: How do we rear domestic animals?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Rearing Selected Animals Management practices: Activity And Practice

Practice for Rearing Selected Animals Management practices:

1. Carry out or describe a practical task for this outcome: describe the management practices in animal rearing. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: carry out routine management practices in animal rearing. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: develop responsibility in the rearing of selected animals.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to rearing selected animals management practices and write two useful sentences about it.

Rearing Selected Animals Management practices: Outcome Check

Outcome: describe the management practices in animal rearing

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Rearing Selected Animals Management practices.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Rearing Selected Animals Management practices: Outcome Check

Outcome: carry out routine management practices in animal rearing

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Rearing Selected Animals Management practices.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Rearing Selected Animals Management practices: Outcome Check

Outcome: develop responsibility in the rearing of selected animals.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Rearing Selected Animals Management practices.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Lesson Opener

Learning area: Livestock Products Obtaining products: wool shearing Obtaining products: Milking

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Livestock Products Obtaining products: wool shearing Obtaining products: Milking.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Livestock Products Obtaining products: wool shearing Obtaining products: Milking.

By the end of this lesson sequence, you should be able to:
- explain the considerations in obtaining selected livestock products
- carry out milking from a dairy animal
- adopt hygiene practices in clean milk production.

Inquiry questions:
- How do we obtain clean products from animals?

Before reading, write one thing you already know and one question you want this lesson to answer.

Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Livestock Products Obtaining products: wool shearing Obtaining products: Milking.

The important idea is Livestock Products Obtaining products: wool shearing Obtaining products: Milking. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Livestock Products Obtaining products: wool shearing Obtaining products: Milking.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Livestock Products Obtaining products: wool shearing Obtaining products: Milking
- Livestock
- Products
- Obtaining
- shearing
- Milking

Explain the idea in your own words before attempting the activities.

Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Worked Example

Practical model for Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Safe animal handling.

Teacher-supervised procedure:
1. Observe the animal's behaviour before moving close.
2. Approach from the side where the animal can see you.
3. Keep hands away from the mouth, horns, hooves, and back legs.
4. Use clean water, feed, bedding, and equipment as instructed.
5. Wash hands and disinfect equipment after the task.

Hazards: kicks, bites, scratches, disease transmission, and stress to the animal.

Quality criteria: calm handling, clean tools, correct restraint if needed, animal welfare, and accurate records of feed, health, or behaviour.

Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Activity

Activity for Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Complete a supervised agriculture procedure card.

Inquiry question: How do we obtain clean products from animals?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Activity And Practice

Practice for Livestock Products Obtaining products: wool shearing Obtaining products: Milking:

1. Carry out or describe a practical task for this outcome: explain the considerations in obtaining selected livestock products. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: carry out milking from a dairy animal. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: adopt hygiene practices in clean milk production.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to livestock products obtaining products: wool shearing obtaining products: milking and write two useful sentences about it.

Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Outcome Check

Outcome: explain the considerations in obtaining selected livestock products

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Livestock Products Obtaining products: wool shearing Obtaining products: Milking.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Outcome Check

Outcome: carry out milking from a dairy animal

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Livestock Products Obtaining products: wool shearing Obtaining products: Milking.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Livestock Products Obtaining products: wool shearing Obtaining products: Milking: Outcome Check

Outcome: adopt hygiene practices in clean milk production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Livestock Products Obtaining products: wool shearing Obtaining products: Milking.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Production: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Agricultural Technologies and Entrepreneurship

In this chapter you will study Agricultural Technologies and Entrepreneurship.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:
- 3.1 Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance practices Safety in use
- 3.2 Management of Agro- byproducts Agricultural by- products and their uses Value addition of agricultural by- products
- 3.3 Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Lesson Opener

Learning area: Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used.

By the end of this lesson sequence, you should be able to:
- identify sources of power used on tools and equipment
- describe powered tools and equipment used for various agricultural tasks
- carry out selected agricultural tasks using powered tools and equipment
- carry out appropriate maintenance practices on selected powered tools and equipment

Inquiry questions:
- How do powered tools and equipment contribute to efficiency of agricultural production operations?

Before reading, write one thing you already know and one question you want this lesson to answer.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used.

The important idea is Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used
- Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance practices Safety in use
- Powered
- Tools
- Equipment
- Sources
- power
- Maintenance

Explain the idea in your own words before attempting the activities.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Worked Example

Practical model for Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Tool selection and maintenance check.

Procedure:
1. Name the agriculture task and choose the correct tool for it.
2. Inspect the handle, blade, joint, edge, or moving part before use.
3. Use the tool with safe spacing and the correct body position.
4. Clean, dry, oil, sharpen, or store the tool as appropriate after use.
5. Record one fault found and the safe maintenance action taken.

Hazards: cuts, loose handles, rust, poor storage, and using the wrong tool.

Rubric: excellent work matches tool to task, checks safety before use, maintains the tool correctly, and records the result.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Activity

Activity for Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Complete a supervised agriculture procedure card.

Inquiry question: How do powered tools and equipment contribute to efficiency of agricultural production operations?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Activity And Practice

Practice for Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used:

1. Carry out or describe a practical task for this outcome: identify sources of power used on tools and equipment. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: describe powered tools and equipment used for various agricultural tasks. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: carry out selected agricultural tasks using powered tools and equipment. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: carry out appropriate maintenance practices on selected powered tools and equipment. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to powered tools and equipment sources of power powered tools and equipment maintenance: sources of power used and write two useful sentences about it.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Outcome Check

Outcome: identify sources of power used on tools and equipment

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Outcome Check

Outcome: describe powered tools and equipment used for various agricultural tasks

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used: Outcome Check

Outcome: carry out selected agricultural tasks using powered tools and equipment

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: sources of power used.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures: Lesson Opener

Learning area: Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures.

By the end of this lesson sequence, you should be able to:
- appreciate the safety measures in the use of powered tools and equipment.

Inquiry questions:
- How do powered tools and equipment contribute to efficiency of agricultural production operations?

Before reading, write one thing you already know and one question you want this lesson to answer.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures.

The important idea is Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures
- Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance practices Safety in use
- Powered
- Tools
- Equipment
- Sources
- power
- Maintenance

Explain the idea in your own words before attempting the activities.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures: Worked Example

Practical model for Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures: Tool selection and maintenance check.

Procedure:
1. Name the agriculture task and choose the correct tool for it.
2. Inspect the handle, blade, joint, edge, or moving part before use.
3. Use the tool with safe spacing and the correct body position.
4. Clean, dry, oil, sharpen, or store the tool as appropriate after use.
5. Record one fault found and the safe maintenance action taken.

Hazards: cuts, loose handles, rust, poor storage, and using the wrong tool.

Rubric: excellent work matches tool to task, checks safety before use, maintains the tool correctly, and records the result.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures: Activity

Activity for Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures: Complete a supervised agriculture procedure card.

Inquiry question: How do powered tools and equipment contribute to efficiency of agricultural production operations?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures: Activity And Practice

Practice for Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures:

1. Carry out or describe a practical task for this outcome: appreciate the safety measures in the use of powered tools and equipment.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to powered tools and equipment sources of power powered tools and equipment maintenance: safety measures and write two useful sentences about it.

Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures: Outcome Check

Outcome: appreciate the safety measures in the use of powered tools and equipment.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Powered Tools and Equipment Sources of power Powered tools and equipment Maintenance: safety measures.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Management of Agro- byproducts Agricultural by- products and their uses Value addition: Lesson Opener

Learning area: Management of Agro- byproducts Agricultural by- products and their uses Value addition

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Management of Agro- byproducts Agricultural by- products and their uses Value addition.

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Management of Agro- byproducts Agricultural by- products and their uses Value addition.

By the end of this lesson sequence, you should be able to:
- explain the uses of various agricultural by-products for economic purposes
- add value to agricultural by- products for economic purposes
- embrace utilisation of agro-byproducts in farming.

Inquiry questions:
- How can agricultural by-products be utilised profitably?

Before reading, write one thing you already know and one question you want this lesson to answer.

Management of Agro- byproducts Agricultural by- products and their uses Value addition: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Management of Agro- byproducts Agricultural by- products and their uses Value addition.

The important idea is Management of Agro- byproducts Agricultural by- products and their uses Value addition. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Management of Agro- byproducts Agricultural by- products and their uses Value addition.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Management of Agro- byproducts Agricultural by- products and their uses Value addition
- Management of Agro- byproducts Agricultural by- products and their uses Value addition of agricultural by- products
- Management
- byproducts
- Agricultural
- products
- their
- Value

Explain the idea in your own words before attempting the activities.

Management of Agro- byproducts Agricultural by- products and their uses Value addition: Worked Example

Practical model for Management of Agro- byproducts Agricultural by- products and their uses Value addition: Produce handling and value-addition plan.

Procedure:
1. Identify the produce and its quality requirements before handling.
2. Sort, clean, grade, dry, package, process, or store it using the method taught.
3. Keep hands, containers, surfaces, and storage areas clean.
4. Label the product or record date, quantity, and condition.
5. Explain how the method reduces loss or improves market value.

Hazards: contamination, cuts, spoilage, wrong packaging, and unsafe storage.

Rubric: excellent work protects hygiene, follows the right method, reduces waste, and explains the value added.

Management of Agro- byproducts Agricultural by- products and their uses Value addition: Activity

Activity for Management of Agro- byproducts Agricultural by- products and their uses Value addition: Complete a supervised agriculture procedure card.

Inquiry question: How can agricultural by-products be utilised profitably?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Management of Agro- byproducts Agricultural by- products and their uses Value addition: Activity And Practice

Practice for Management of Agro- byproducts Agricultural by- products and their uses Value addition:

1. Carry out or describe a practical task for this outcome: explain the uses of various agricultural by-products for economic purposes. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: add value to agricultural by- products for economic purposes. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: embrace utilisation of agro-byproducts in farming.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to management of agro- byproducts agricultural by- products and their uses value addition and write two useful sentences about it.

Management of Agro- byproducts Agricultural by- products and their uses Value addition: Outcome Check

Outcome: explain the uses of various agricultural by-products for economic purposes

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Management of Agro- byproducts Agricultural by- products and their uses Value addition.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Management of Agro- byproducts Agricultural by- products and their uses Value addition: Outcome Check

Outcome: add value to agricultural by- products for economic purposes

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Management of Agro- byproducts Agricultural by- products and their uses Value addition.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Management of Agro- byproducts Agricultural by- products and their uses Value addition: Outcome Check

Outcome: embrace utilisation of agro-byproducts in farming.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Management of Agro- byproducts Agricultural by- products and their uses Value addition.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Lesson Opener

Learning area: Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts..

Curriculum link: This lesson follows the Grade 11 curriculum sequence and focuses on Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts..

By the end of this lesson sequence, you should be able to:
- describe documents used in agri-business transaction
- prepare documents used in agri-business transaction
- acknowledge the role of agri-business accounts in agricultural entrepreneurship.

Inquiry questions:
- How is documentation of agri-business transactions done?

Before reading, write one thing you already know and one question you want this lesson to answer.

Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts..

The important idea is Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

For Grade 11, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts..
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:
- Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.
- business
- Accounts
- Documents
- agri-business
- transactions
- agribusiness
- accounts.

Explain the idea in your own words before attempting the activities.

Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Worked Example

Practical model for Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Simple farm-record or enterprise check.

Procedure:
1. Name the activity, such as planting, feeding, harvesting, selling, or storing.
2. Record date, item, quantity, cost, income, loss, or observation.
3. Calculate one useful figure, such as total cost, total sales, profit, loss, or stock remaining.
4. Explain one decision the record supports.
5. Store the record neatly for comparison with the next activity.

Hazards: wrong measurements, missing entries, unsafe storage, and confusing income with profit.

Rubric: excellent work has accurate entries, correct calculation, a realistic decision, and neat records.

Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Activity

Activity for Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Complete a supervised agriculture procedure card.

Inquiry question: How is documentation of agri-business transactions done?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:
- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Activity And Practice

Practice for Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.:

1. Carry out or describe a practical task for this outcome: describe documents used in agri-business transaction. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: prepare documents used in agri-business transaction. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: acknowledge the role of agri-business accounts in agricultural entrepreneurship.. List materials, steps, safety checks, and how you will judge quality.

Correction check:
- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to agri- business accounts documents in agri-business transactions role of agribusiness accounts. and write two useful sentences about it.

Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Outcome Check

Outcome: describe documents used in agri-business transaction

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts..
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Outcome Check

Outcome: prepare documents used in agri-business transaction

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts..
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: Outcome Check

Outcome: acknowledge the role of agri-business accounts in agricultural entrepreneurship.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts..
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agricultural Technologies and Entrepreneurship: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Glossary

Use this glossary to revise important words in Agriculture.

- Inorganic fertilisers Soil nutrients Categories of fertilisers Soil sampling Soil pH amendment: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Water conservation Methods of irrigation Importance of irrigation: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Vegetative propagation Methods of vegetative propagation Use of vegetative propagation: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Crop Production: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Crop Protection: Pest control Categories of pests Control of pests Economic importance of pests: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Crop Production Structures in crop production Construction materials Construction of structures: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Animal Production: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Rearing Selected Animals Management practices: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Livestock Products Obtaining products: wool shearing Obtaining products: Milking: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Agricultural Technologies and Entrepreneurship: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Agri- business Accounts Documents in agri-business transactions Role of agribusiness accounts.: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.

Add five more words from your lessons, then write a meaning and one correct example sentence or worked example for each word.

Answer And Teacher Notes

Most questions in this book are open practice tasks. A good answer should:

- respond to the exact question asked;
- use correct vocabulary from Agriculture;
- include a clear example, reason, working, observation, or evidence;
- be neat enough for another learner to follow;
- show correction after feedback.

For calculations, check each step. For language work, check spelling, punctuation, grammar, and meaning. For practical subjects, check safety, materials, observations, and conclusion.

Answer-key guide: Social Studies answers should name the place, people, evidence, meaning, and responsible action. Agriculture answers should name tools, sequence, safety, observation, and care for living things. Creative Arts answers should show planning, technique, safety, improvement, and explanation of choices.

Rubric for self-checking:
- 4: The answer addresses the exact question, uses subject vocabulary correctly, gives evidence or working, and includes a useful check or correction.
- 3: The main answer is correct but one part needs stronger evidence, clearer working, better labels, or a fuller explanation.
- 2: Some understanding is shown, but the answer is incomplete, too general, or hard to follow.
- 1: The answer is guessed, off topic, or missing the evidence needed for the task.

Correction routine: reread the command word, underline the key data or evidence, improve one weak step, and write one sentence explaining why the final answer is reasonable.

Final Project

Create a final project for Agriculture.

Your project must include:
1. A clear title.
2. A real-life problem or question.
3. Evidence from at least three lessons.
4. A drawing, table, map, chart, model, dialogue, performance plan, artwork note, or worked solution.
5. A short reflection explaining what you learned.

Present your project to a classmate, teacher, parent, or study group.